award

Association for Water and Rural Development

CROSS-CATCHMENT COLLECTIVE

Catchment Wisdom Framework in practice

Case study for the CCC applied across six rivers of the Lowveld

Submitted by

River of the Lowveld Collective

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This report

This report demonstrates the Catchment Wisdom Framework in practice for the Rivers of the Lowveld, drawing primarily on the Shared Rivers Initiative (SRI) evidence base. It provides short profiles for six river systems and shows how the full DIKIW pathway (Data → Information → Knowledge → Insight → Wisdom) can convert technical and social evidence into collective, adoptable guidance—what to do, why, for whom, and under what conditions. The report also includes a guide for cross-catchment wisdom and embedded placeholders for the key maps/diagrams needed to communicate findings by the CCC.

LIST OF Abbreviations & Acronyms

BBR	Bushbuckridge
BBBEE	Broad Based Black Economic Empowerment
BHNR	Basic Human Needs Reserve
CCAWs	Co-ordinating Committees on Agricultural Water
CIB	Crocodile Irrigation Board
CMA	Catchment Management Agency
CMC	Catchment Management Committee
CME	Compliance Monitoring and Enforcement
CMF	Catchment Management Forum
CMIB	Crocodile Major Irrigation Board
CMS	Catchment Management Strategy
Cofomosa	Committee for Facilitation of Agriculture between Mozambique and South Africa
CPA	Common Property Association
CRS	Controlled Release Scheme
DAFF	Department of Agriculture, Forestry and Fisheries
DARDLA	Agriculture, Rural Development and Land Affairs, Mpumalanga
DEA	Department of Environmental Affairs (previously DEAT and included tourism)
DFA	Development Facilitation Act (65 of 1995)
DEDET	Department Economic Development, Environment, and Tourism
DM	District Municipality
DMR	Department of Mineral Resources (previously DME)
DNA	National Directorate for Water Affairs (Mozambique)
DoA	Department of Agriculture (now DAFF)
DPLG	Department of Provincial and Local Government
DSS	Decision Support System
DWA	Department of Water Affairs (previously DWAF and included forestry)
EIA	Environmental Impact Assessment
EF	Environmental Flows
EIP/EMP	Environmental Implementation Plan/Environmental Management Plan
EMPR	Environmental Management Programme Report
EMPs	Environmental Management Plans
EWRs	Environmental water requirements
ER	Ecological Reserve (referred to as the Reserve)
FBW	Free Basic Water
FSC	Forest Stewardship Commission
FSE	Federation for Sustainable Environment
GA	General Authorisation
GEAR	Growth, Employment and Redistribution (A Macroeconomic Strategy for South Africa)
GGP	Gross Geographic Product
GIS	Geographic Information System
GLTFCA	Great Limpopo Transfrontier Conservation Area
GLWP	Greater Letaba Water Project
GLWUA	Groot Letaba Water User Association
HDI	Historically Disadvantaged Individual
IAPs	Interested and Affected Parties
IBT	Inter-Basin Transfer
ICMA	Inkomati Catchment Management Agency
IDP	Integrated Development Plan
IEM	Integrated Environmental Management
IFR	In-stream flow requirement
IIMA	The Interim Inco-Maputo Agreement
IMC	Irrigation Management Committees
IRDS	The Integrated Rural Development Strategy
ISO	International standards
ISOTG	Incomati System Task Group
ISP	Internal Strategic Perspective

IWMP	Integrated Waste Management Plan
IWRM	Integrated Water Resources Management
IWRMP	Integrated Water Resource Management Plan
JPTC	Joint Permanent Technical Committee which became the JWC
JWC	Joint Water Commission
K2C	Kruger to Canyons Biosphere Reserve
KIB	Komati Irrigation Board
KJOF	Komati Joint Operations Forum
kl/hh/m	kiloliters per household per month
KNP	Kruger National Park
KOBWA	The Komati Basin Water Authority
LBPTC	Limpopo Basin Permanent Technical Committee
LG / LoGov	Local Government
LIB	Loskop Irrigation Board
LM	Local municipality
LRAD	Redistribution for Agricultural Development policy
LRO	Limpopo Regional Office
LUMS	Land Use Management Systems
M&E	monitoring and enforcement
MIB	Major Irrigation Board
MIG	municipal infrastructure grant
MNRE	Swaziland Ministry of Natural Resources and Energy
MOU	Memorandum of Understanding
MSA	Municipal Systems Act (No 32 of 2000)
MSPs	multiple stakeholder platforms
MTPA	Mpumalanga Tourism and Parks Agency
NEMA	National Environmental Management Act (No. 107 of 1998)
NEPAD	New Partnership for Africa's Development
NGO	Non-Governmental Organisation
NLP	National Land Care Programme
NPS	National Pricing Strategy
NWA	National Water Act (No. 36 of 1998)
NWRCS	National Water Resource Classification System
NWRIA	National Water Resources Infrastructure Agency
NWRMC	National Water Resource Monitoring Committee
NWRS	National Water Resource Strategy
OLLI	Olifants Letaba Luvuvhu Incomati (low flows committee)
ORF	Olifants River Forum
ORWRDP	Olifants River Water Resource Development Project
PES	Present Ecological Status
PGDS	Provincial Growth and Development Strategy
PRIMA	Progressive Realisation of the IncoMaputo Agreement
RA	Resilience Alliance
RBA	River Basin Authorities
RDM	Resource Directed Measures
RO	Regional Office
RQO	Resource Quality Objectives
SADC	Southern African Development Community
SALGA	South African Local Government Association
SAM	Strategic Adaptive Management
SANParks	South African National Parks
SDC	Source Directed Controls
SDFs	Spatial Development Frameworks
SFRA	Stream Flow Reduction Activities
SHEQ	Safety, Health, Environment and Quality Management
SRI	Shared Rivers Initiative
STEEP	Social, Technological, Environmental, Economic and Political
TA	Traditional Authority

TPTC	Tripartite Permanent Technical Committee with Mozambique and Swaziland
WAP	Water Allocation Plan
WARMS	Water Authorisation and Registration Management System
WAS	Water Allocation Schedule
WC/WDM	Water Conservation and Water Demand Management
WDCS	Waste Discharge Charge System
WDM	Water Demand Management
WfW	Working for Water
WMA	Water Management Area
WMI	Water Management Institution
WRC	Water Research Commission
WRSAS	Water Resource Situation Assessment Study
WSA	Water Services Act (No. 108 of 1997)
WSA / WSP	Water Services Authorities or Providers
WSAU	Water Services Authority
WSDP	Water Services Development Plan
WSI	Water Services Institute
WSP	Water Service Providers
WUA	Water User Association

Contents

List of Figures

List of tables

1. Introduction: Catchment Wisdom for river basin management

Catchment Wisdom is the Cross-Catchment Collective's way of turning what we already track (data, monitoring, stories, reflections, and evidence) into practical, shared guidance that changes decisions. It is designed to sit *on top of* existing monitoring, evaluation, research and learning (MERL) systems—not as an extra burden, but as the synthesis layer that ensures learning becomes action. The starting point is simple: if we only report outputs and indicators, we miss the "so what" that determines whether water governance and ecosystem protection actually improve.

At the core of the Catchment Wisdom framework is the **DIKIW pathway**—**Data → Information → Knowledge → Insight → Wisdom**. DIKIW helps us name the full journey from raw evidence to collective guidance. In many programmes, MERL is strong at the front of the pipeline (data and reporting), but weaker at the back end (insight and wisdom). Catchment Wisdom fills that gap by creating the structured reflection, collective interpretation, and decision-linkages needed to move from "what happened" to "what should we do next—and why".

Catchment Wisdom is also grounded in **evidence-based planning and adaptive management**. Evidence-based planning asks: what does the best available evidence suggest we should prioritise? Adaptive management asks: how do we act, learn, and adjust in conditions of uncertainty and change? The Shared Rivers Initiative (SRI) work repeatedly highlights that water management in South Africa is increasingly decentralised and participatory—placing a high level of responsibility on local stakeholders for decision-making in evolving contexts, which *requires* learning, adaptation and response over time.[1] SRI further frames **social learning** as a process strategy for supporting the emergence of innovation, with practical preconditions such as a sense of urgency, felt interdependence, stakeholder platforms, facilitation, leadership, and reflection "built in from the start".[2] Catchment Wisdom operationalises these conditions as an explicit, repeatable practice across catchments.

This report therefore treats the **MERL system as the platform**: it provides the routine machinery to collect, store and share data and information. Catchment Wisdom then uses that platform to surface what MERL cannot easily capture: patterns, lived experience, tacit knowledge, and governance "blockages" that determine whether plans are implementable. A clear example from the SRI evidence base is the difficulty of *operationalising and progressively realising the Ecological Reserve* in real institutional settings. In the Letaba context, the Reserve is shown to be constrained by planning gaps, allocation realities, and repeated setbacks in determining ecological water requirements; meeting Reserve requirements can become unlikely without changes such as compulsory licensing and more integrated planning.[3] Across systems, SRI analysis also points to recurring reasons for non-compliance that sit squarely in the "wisdom" space: weak integration between water resources and water services (including municipalities and, in some areas, mining), poor understanding and usability of Reserve information, lack of clear local leadership, unlawful water uses, and limited legal/regulatory skills.[4] These are not just "findings"—they are *guidance about what must change* for implementation to be realistic.

A second example is the Sand/Sabie-Sand system, where Reserve operating rules and progressive realisation are framed as sufficiently contested that legal action is considered—illustrating how governance, law, institutional accountability and social power can become the determining levers of water outcomes.[5] This kind of institutional learning is essential to catchment management, but it often does not appear in standard MERL output tables. Catchment Wisdom is the mechanism that captures these lessons in a way that is actionable and transferable.

a) What "wisdom" is in this report

In this report, wisdom is defined as:
collectively agreed, experience-informed guidance that shapes future decisions, priorities, and ways of working (not just a lesson learned, and not just a recommendation written by one team). Each catchment and sub-catchment will define its own "wisdom statements" through structured reflection—while the CCC synthesises what travels across contexts.

What wisdom should reasonably affect (so it doesn't become navel-gazing)

Catchment Wisdom is only credible if it changes something. Practically, the "wisdom effect" should show up in at least four places:

- **Planning and investment choices** (what gets funded, maintained, enforced, and monitored)
- **Governance practice** (who is involved, how decisions are made, how accountability works)
- **Implementation feasibility** (removing recurring blockers like skills gaps, institutional fragmentation, unclear local leadership, weak compliance systems)
- **Research and policy influence** (e.g., issues and propositions taken to the Water Research Commission and partners for targeted research, method development, or policy reform)[1][4]

Mobilising wisdom through the CCC

The CCC's role is to help ensure that wisdom moves between catchments—turning isolated lessons into a shared asset. That happens through repeatable processes: cross-catchment learning records, facilitated reflection, synthesis briefs, and structured engagements with research and policy actors. The point is not to create more reporting. The point is to create *better decisions*.

Common datapoints: what cross-catchment evidence offers that single projects cannot

A core intention of the CCC is that common datapoints across multiple catchments unlock a systems view that no single catchment can see alone. Across the six catchments, the shared evidence base should enable:

- Trend analysis across space and time (e.g., recurring flow–quality–governance patterns)
- Comparative understanding of what works *where* and *why* (context conditions, enabling factors, failure modes)
- Early warning signals that only emerge at scale (systemic risks, institutional bottlenecks, cumulative land-use impacts)

In practical terms, the shared evidence tends to cluster around: **ecosystem protections and environmental flows**, **hydrological studies and monitoring**, **land use and water quality impacts**, and **governance and compliance dynamics**—including the "soft infrastructure" of leadership, institutional coordination, and legal/regulatory capability that determines whether technically sound measures are implemented.[3][4]

2. Catchment Wisdom: some basics

Catchment Wisdom is the collective, experience-informed guidance that partners agree on to shape future decisions, policies and practices across catchments. It is not a report, a dataset, or a set of "nice insights". It is the point where evidence and lived realities are translated into shared direction—the practical guidance that influences what partners prioritise, what institutions adopt, and what changes in day-to-day practice.

Catchment Wisdom emerges when the full **DIKIW journey**—**Data → Information → Knowledge → Insight → Wisdom**—is **co-created, validated and applied at catchment scale.** That "catchment scale" clause matters: it signals that wisdom is not just what one project team believes, but what a broader set of stakeholders can recognise, test, and use.

What makes Wisdom different from "learning"

Many programmes can point to lessons learned. Catchment Wisdom goes further. It has four distinguishing features:

1. **It is collective and negotiated**
   Wisdom is built through shared interpretation across stakeholders (communities, water institutions, municipalities, researchers, traditional authorities, water users). If it is not collectively held, it is not yet wisdom.
2. **It is evidence-backed and experience-informed**
   Wisdom sits on both "hard" and "soft" evidence: monitoring data and reports, plus lived

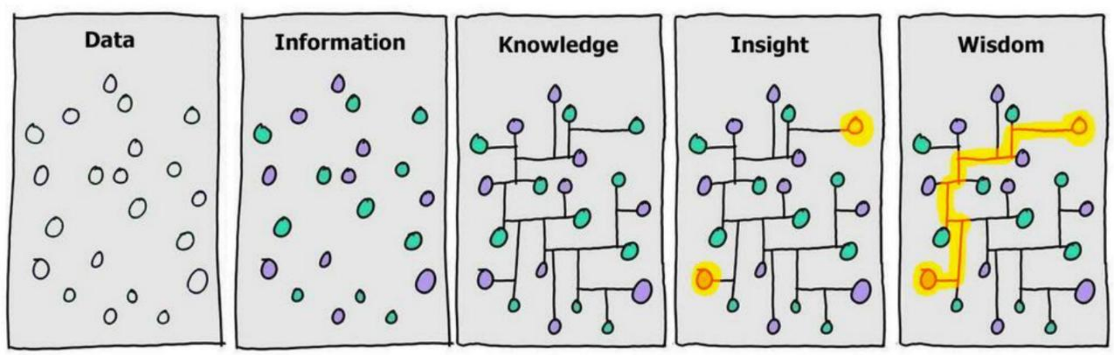
experience, local knowledge and institutional memory—especially around governance constraints and feasibility.

3. **It is decision-linked**
   Wisdom must be expressed in ways that can be adopted: rules of thumb, protocols, co-management agreements, planning commitments, prioritisation criteria, or advocacy messages.

4. **It is revisable**
   Catchment conditions change. Wisdom is not a permanent conclusion; it is guidance that is reviewed as new evidence and new realities emerge.

Catchment Wisdom = The *collective, experience-informed guidance* that partners agree upon to shape future decisions, policies, and practices across catchments.

Catchment Wisdom is when the "aha!" moments from data, science and lived experience are crystallised into shared norms, agreements or practices that shape collective practices for managing land and water on a catchment scale.



Information Isn't Power, illustration by David Somerville.

It goes beyond technical knowledge by integrating scientific evidence, local knowledge, lived experience, and institutional memory. Catchment Wisdom emerges when:

- Data (raw observations, measurements, records) are systematically gathered.
- Information is generated by processing and contextualising data for relevance.
- Knowledge develops when information is interpreted, compared, and validated by different actors against specific contexts.
- Insight arises through reflection, dialogue, and application, revealing patterns, drivers, and trade-offs.

Through co-creation and validation at catchment scale, this iterative process produces guidance that is:

- Collective – shaped by multiple stakeholders (communities, government, researchers, private sector).
- Contextual – grounded in the unique social, ecological, and hydrological realities of the catchment.
- Action-oriented – translated into practical pathways for improved governance, sustainable use, and resilience.
- Evolving – continuously refined as new data, experiences, and innovations emerge.

In practice, *Catchment Wisdom* represents a shared compass: it helps partners align strategies, anticipate risks, and act collectively, ensuring that water, land, and livelihoods are managed in ways that balance equity, resilience, and sustainability.

3. Guiding Principles

1. Learning first: MERL is designed to generate insight, not just track compliance.
2. Participatory: Communities, partners, and stakeholders shape what is learned, how it is interpreted, and how findings are applied.
3. Adaptive: Findings inform ongoing adaptation, not just end-of-project reporting.
4. Systems-oriented: Focus on interactions, patterns, and change across catchments rather than isolated outputs.
5. Equity and inclusion: Diverse voices, especially marginalised groups, are central to sense-making.

4. The Wisdom Journey

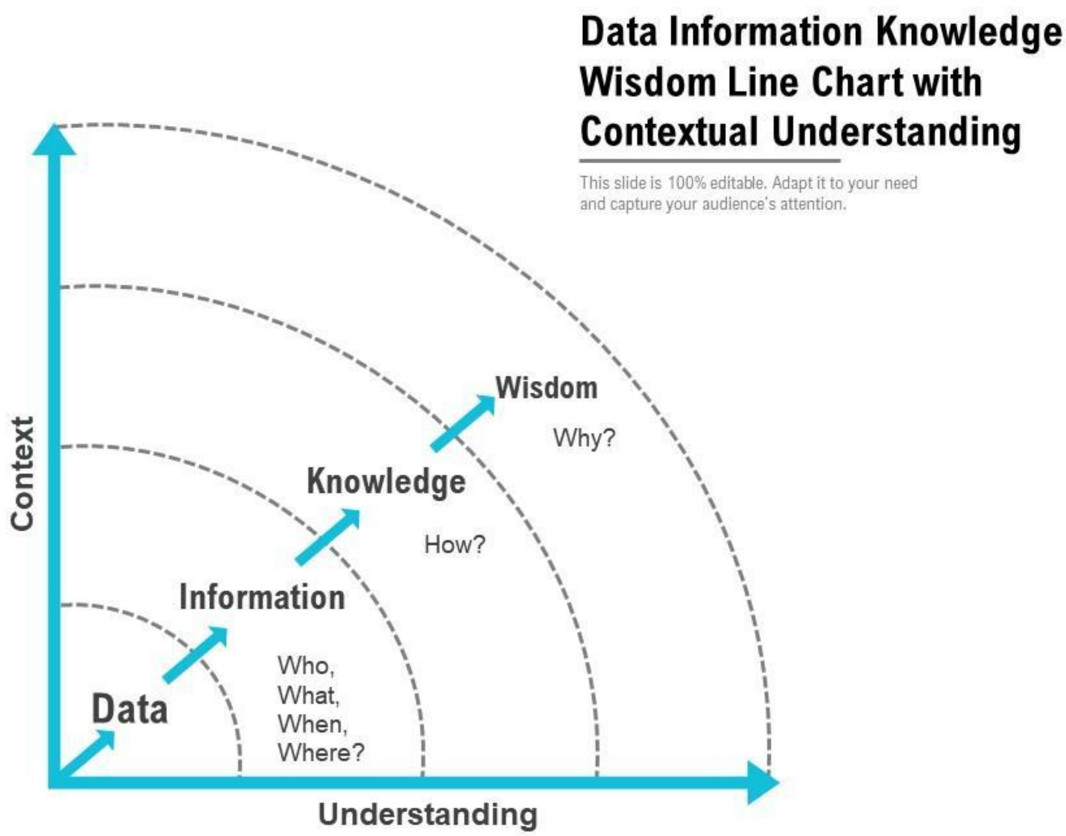
We have applied the full DIKIW pathway (Data → Information → Knowledge → Insight → Wisdom) to six catchments of the Lowveld Rivers (see later). In that sense it is the learning spine of the Catchment Wisdom Journey that we have embarked upon.

In this chapter, DIKIW is used to move from river system evidence (hydrology, water quality, water use, institutional practice and lived experience) to shared, experience-informed guidance that can be embedded in operating rules, restriction protocols, planning cycles and forum agendas. DIKIW is treated as iterative: as conditions change and actions are tried, new data and reflection update guidance.

Key points of the Wisdom Journey

- Data: raw measurements, observations and stories (e.g., flow gauges, dam releases, abstraction volumes, incident logs, community minutes).
- Information: data cleaned, visualised and contextualised (e.g., compliance graphs, maps, dashboards, water-balance summaries).
- Knowledge: shared interpretation and validation among stakeholders (e.g., agreed explanations of why compliance is failing in a reach).

- Insight: collective reflection that surfaces drivers, feedbacks and leverage points (e.g., operating-rule adherence, incentive structures, institutional lags).
- Wisdom: agreed guidance that shapes action and can be reviewed over time (e.g., revised operating rules, forum-backed restriction plans, monitoring/enforcement priorities).



Step	Process	Example Output
1. Data	Field measurements, observations, stories	Rainfall logs, community minutes
2. Information	Data cleaned, visualised, contextualised	Graphs, maps, dashboards
3. Knowledge	Shared interpretation and validation	“Solar pumps improved access in X area”
4. Insight	Collective reflection revealing drivers	“Local leadership reduces system failures”

5. Wisdom	Agreed guidance shaping action	"Integrate traditional leaders in co-management"

How this method makes the DIKIW journey visible

The Catchment Wisdom approach makes learning traceable and useful by doing three things:

1. **Documenting the transformation**
   Partners show *how* raw data became higher-order understanding. This prevents the common gap where "we collected lots of data" is assumed to equal "we now know what to do".
2. **Stitching individual learning into system-wide wisdom**
   Collective workshops and learning labs help connect what each site is seeing into broader patterns. This is where cross-catchment value is created: trends, recurring drivers, and transferable practices become visible.
3. **Producing accessible outputs**
   Wisdom outputs are designed for use—clear statements, briefs, maps, dashboards, and learning summaries that show the value of collective evidence (not just isolated monitoring results). The goal is uptake: planning, coordination, and policy influence.

b) Details of the Pathways to Wisdom

The Pathways to Wisdom (also called the Wisdom Journey) describes the practical steps that move evidence into action-ready guidance. Each step has a clear purpose and a clear output.

Data — Raw facts from the field

The unprocessed data —numbers, observations, stories, and records. Fragmented inputs that show "what is happening" but not yet "what it means."

Data

- Collected by communities, partners, researchers, and technical tools.
- Examples: rainfall readings, water quality tests, farmer practices, community meeting notes.

Who collects it: Communities, implementing partners, researchers, technical teams, and tools such as sensors, sampling kits, and monitoring platforms.

Key risk at this step: Data overload or "data without purpose" that never informs decisions.

Examples:

- Rainfall and flow readings

- Water quality test results
- Pump functionality logs and maintenance notes
- Records of farmer practices (mulching, irrigation schedules)
- Community meeting minutes, voice notes, conflict reports
- Photos/video of infrastructure failures or restoration work

Information — Organised data

Data that has been cleaned, structured, visualised and contextualised so it can be shared and discussed. Evidence that is accessible to a wider group and can support common reference points in discussions.

It involves typical processing**:** sorting, summarising, mapping, basic analysis, comparison across sites/time.

Key risk at this step: Information becomes "outputs for reporting" rather than tools for learning and decision-making.

Information (organised data)

- Data cleaned, sorted, visualised, and compared.
- Examples: maps of water access points, graphs of rainfall trends, participation records

Examples:

- Maps of water points, supply reliability, or pollution hotspots
- Graphs of rainfall trends, groundwater levels, or service functionality
- Dashboards summarising water access or system breakdown frequency
- Participation records (who attended, who spoke, who is missing)
- Simple seasonal summaries (wet vs dry season patterns)

Knowledge — Shared understanding and interpretation

A common interpretation built through collective sense-making. This is where evidence is linked to context, lived experience, and local knowledge. A shared story of cause and effect—what the evidence suggests is happening and why.

Generally it is produced by dialogues, validation sessions, triangulation across data sources and stakeholders.

Key risk at this step: Dominant voices define "knowledge" while other stakeholders' experience is excluded.

Examples:

- “Water access improved in villages where solar pumps were installed.”
- “Breakdowns cluster in systems without a clear maintenance role and budget.”
- “Water quality deterioration correlates with upstream land-use change and sewage failures.”

Knowledge (shared understanding or interpretation)

- Collective sense-making by partners.
- Evidence linked to lived experience and local knowledge.
- Example: “Water access improved in villages where solar pumps were installed.”

Insight — New perspectives and implications

The deeper learning that emerges when knowledge is compared across stakeholders, sites, and time—revealing drivers, constraints, and trade-offs. Clear implications for strategy—what needs to change, what trade-offs must be managed, and what enabling conditions are required.

Generally produced by reflection across cases, cross-catchment comparison, structured learning labs, and facilitated “so what” discussions.

Key risk at this step: Insight stays in workshop notes and never becomes guidance that institutions adopt.

Examples:

- “Strong local leadership reduces infrastructure breakdowns because response is quicker and accountability is clearer.”
- “Catchment-wide issues (e.g., drought rules, pollution response) cannot be solved village-by-village; they require coordination across users and institutions.”
- “Reserve determination is technically sound but often fails in practice because it is too slow, too complex, and too costly to operationalise—and the legal pathway is too specialised for routine enforcement.”

Insight (new perspectives and implications)

- Learning across stakeholders reveals patterns, drivers, constraints, and trade-offs.
- Examples: “Strong local leadership reduces breakdowns” or “Catchment-wide issues need inter-village coordination.”

- "Municipalities behave differently in communal areas than commercial/titled land contexts; governance interfaces must be designed accordingly."

Wisdom — Catchment-level guidance for action

Insight translated into **adoptable action**—strategies, agreements, rules, norms, and policy-relevant recommendations that shape how decisions are made and resources allocated. Shared guidance that shapes decisions, coordination, investment, and advocacy (e.g., what is escalated to the Water Research Commission, for example, and why).

Expressed as concise "wisdom statements" with owners, conditions, and review triggers.

Key risk at this step: "Navel-gazing" — producing wisdom statements without adoption pathways, owners, or linkage to real decision windows.

Examples of wisdom outputs:

- **Co-management and governance:** "Co-management agreements must include traditional leaders and clarify roles between municipalities, water institutions and community structures."
- **Service reliability:** "Maintenance systems must have a funded role, a response time standard, and trained local operators to reduce repeat failures."
- **Water-smart livelihoods:** "Scaling water-smart agriculture requires enterprise support (inputs, markets, finance), not only technical training."
- **Reserve operationalisation:** "Reserve determination must be simplified into operational rules and decision triggers that local institutions can implement—not left as technical studies."
- **Cross-catchment learning:** "Minimum common datasets and shared learning records are required to detect trends and guide regional advocacy."

Wisdom (catchment-level guidance for action)

- Insights translated into strategies, agreements, norms, or policies.
- Applied across villages and catchments; linked to decision windows and ownership.
- Examples: "Co-management agreements must include traditional leaders."; "Scaling water-smart agriculture requires local enterprise support."

A practical quality TEST: is it really Catchment Wisdom?

A statement qualifies as Catchment Wisdom if it can answer all five questions:

1. **What should change?** (guidance and what it aims to contribute to)
2. **Why?** (evidence + experience for why it should be adopted)
3. **For whom?** (equity and risk issues explained)
4. **Who must adopt it?** (institution/actor identified and consulted)
5. **When will we review and reflect on it?** (trigger/date)

This test keeps the Wisdom Journey action-oriented and prevents the process from becoming reflective and disconnected. Use this TEST to prevent "navel-gazing". Each wisdom statement should be linked to a real decision window (planning, budgeting, bylaws, operations) and include minimum enabling conditions (data, capacity, finance, and governance roles).

5. An example of insights to learning… at a catchment scale

INTERNATIONAL
WATER POWER
AND DAM CONSTRUCTION

Grand Maison Dam in France. Photo courtesy of EDF Hydro DTG

November 2025

Condition monitoring

Next-generation dam monitoring insights

Steven Thurgood and Lidija Špiranec from Leica Geosystems, part of Hexagon, explore how the right monitoring solution enables dam operators to ensure safety, meet regulatory requirements, and use data to validate and improve engineering designs.

Dam monitoring provides early warnings of structural issues, helping operators and owners manage risks and meet regulatory standards. While monitoring technology has advanced considerably, truly effective outcomes depend on how well dam monitoring systems are designed, installed, configured, operated, and maintained.

As providers of high-precision monitoring systems for dams, Leica Geosystems, part of Hexagon, has supported dam owners and operators worldwide, providing the technology they need to ensure structural health and safety, from robotic total stations and GNSS (Global Navigation Satellite System) sensors to the software that automates measurements and brings the data together.

Here, we highlight three frequent pitfalls in dam monitoring and how to overcome them.

Understanding and configuring the data

The most common problem we see in dam monitoring systems is incorrect use of technology. For example, we recently saw a case where a total station installed to measure the movement of a dam was also moving with the structure, without any stable external reference points. This meant the operators couldn't distinguish between the movement of the instrument and the movement of the dam by looking at the data, making it less relevant for deformation analysis. Yet, even with a perfectly installed system, the biggest risk is that the data is not actively reviewed by experienced personnel. A monitoring system will be ineffective if data is either not reviewed or poorly analysed.

Tailoring the solution to both the site conditions and primary stakeholders is one solution. By using one integrated system, for example an automated system of geotechnical sensors and GNSS, operators can receive a real-time structural health check on critical assets. A central software platform integrates, analyses, and visualises the data from these varied sources into a single, understandable interface.

Smart integrated systems can send alerts when movements exceed certain thresholds, keeping stakeholders up to date with the information they need to act swiftly. As required, expert land surveyors can then provide periodic surveys using a total station to gather more detailed spatial data across the wider structure. This hybrid approach allows for a comprehensive, cost-effective, and risk-appropriate monitoring plan.

Outdated technology

Another common issue is outdated technology. Many dams still rely on systems installed decades ago, which are often only replaced after disasters or when enforced by new legislation.

For example, one dam we visited recently had been relying on monitoring equipment installed in the 1970s. Parts kept failing as time went on, and for the last 15 years, the installed system has become ineffectual. The introduction of new legislation – due to increased risk of natural disasters and heavy rainfall in the region – meant the dam operators were obliged to update their monitoring solution as a matter of public safety.

While outdated or poorly maintained technology can result in incomplete data, modern systems capture comprehensive data to give the full picture of the dam's behaviour over time. Fully automated and permanent installations provide continuous data without human intervention, an often-necessary approach for high-risk or remote sites.

Robotic total stations, a key technology in such systems, automatically measure prisms installed on the dam, delivering high-accuracy 3D coordinates to track surface displacements. Additionally, GNSS technology offers absolute 3D position data, which is vital for monitoring remote assets or confirming the stability of any reference network used.

A wide array of geotechnical sensors is typically integrated to understand the structural health of the dam and adjacent assets. These include piezometers to measure water pressure, tiltmeters for rotation, strain gauges for load and stress, and crack gauges for expansion and contraction. Many can be deployed as part of a wireless mesh network, simplifying installation. To ensure reliability, these systems can use solar power with battery backup should the direct supply fail.

The East Bay Municipal Utility District (EBMUD) in California demonstrates the benefit of a modern system. After severe storms, EBMUD installed one of the nation's most advanced automated GNSS systems at its Pardee and Camanche reservoirs. The monitoring solution reports crest elevations remotely, helping EBMUD reduce the time needed for surveys. Its value was proven when, after a 4.4 magnitude earthquake, engineers were able to log in to the system remotely and immediately confirm that there had been no significant movement on the dam. Moreover, the automated system helps them meet state monitoring requirements since it can be more easily tied into state-wide emergency and seismic monitoring systems.

Staff & resource optimisation

A significant challenge for dam owners is the need for high-quality monitoring data, which often requires advanced skills, substantial resources, and careful planning. With limited access to large teams for frequent manual surveys – particularly at remote locations – automation has become an essential solution to ensure dams remain safe and compliant. These challenges arise from a combination of factors, including resource constraints, specialised expertise, and the inherent risks involved, rather than shortcomings on the part of dam operators.

One example of this is Enel Green Power, a multinational renewable energy company. Four *of their hydroelectric plants in Italy and Spain were struggling with scheduling and availability to obtain manual deformation measurements, leading to a lack of continuity. So, they installed an automated GNSS monitoring system, which now provides continuous, 24/7 data on the structural health of the dams, eliminating the need for costly and time-consuming longdistance travel for manual checks.*

It allowed them to correlate geodetic data with geotechnical sensors, creating a more detailed and complete description of any deformation. Importantly, the new system provides data about the infrastructure's condition all the time, not just when survey crews are available.

Learning from the data

Monitoring systems, however, provide data that does more than mitigate risks – they can also shape the future of dam construction. Automated systems provide rich datasets that can validate the original design assumptions or highlight areas where future designs could be improved, saving long-term costs and increasing safety.

This is because the large amount of data collected by automated systems allows engineers to understand the true, dynamic behaviour of a dam under various real-world loads. Viewing monitoring as a design validation tool facilitates safer, more efficient, and more resilient dams. Ultimately, effective monitoring enables operators and regulators to dynamically understand and manage risk, keep dams safe, and ensure regulatory compliance

The Wisdom Journey applied to Rivers of the Lowveld

In this section we apply the Wisdom Framework and Wisdom Journey to work conducted by AWARD and partners in the catchments of six lowveld rivers over a period of more than 15 years, under multiple projects.

Figure 1 Rivers of the lowveld are subject to highly variable flow regimes based on rainfall patters

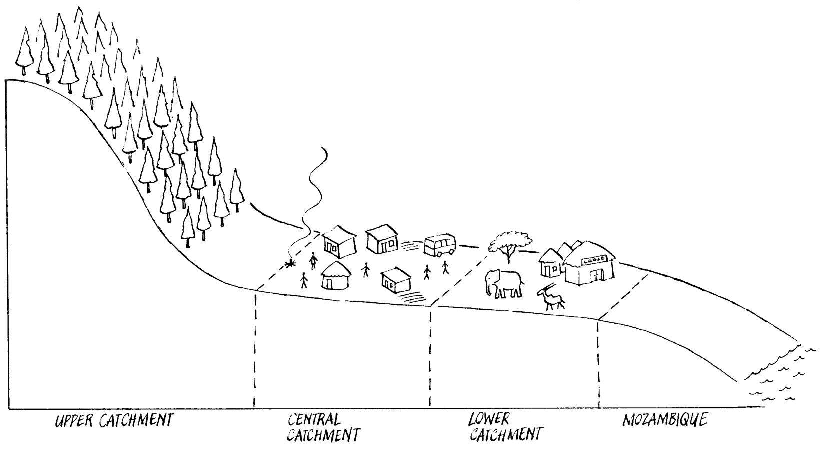


Figure 2 The topography of all six rivers of the Lowveld is similar. Wiull all flowing through the Kruger National Park and into Mozambique

6. Study area overview: six Lowveld rivers

The six river systems used in this case study—Luvuvhu, Letaba, Olifants, Sabie–Sand, Crocodile and Komati—span Limpopo and Inkomati WMAs and feed into the Limpopo and Inkomati international basins. They traverse steep escarpment headwaters into lowland plains, with strong seasonality and high interannual variability. Across the six, the dominant pressures include irrigation, municipal supply expansion, mining/industry and (in places) power-sector pumping incentives, alongside high-value downstream ecosystem services. The recurring implementation challenge is translating the Ecological Reserve from determinations and plans into day- to- day operational practice under competition, institutional complexity and enforcement constraints.

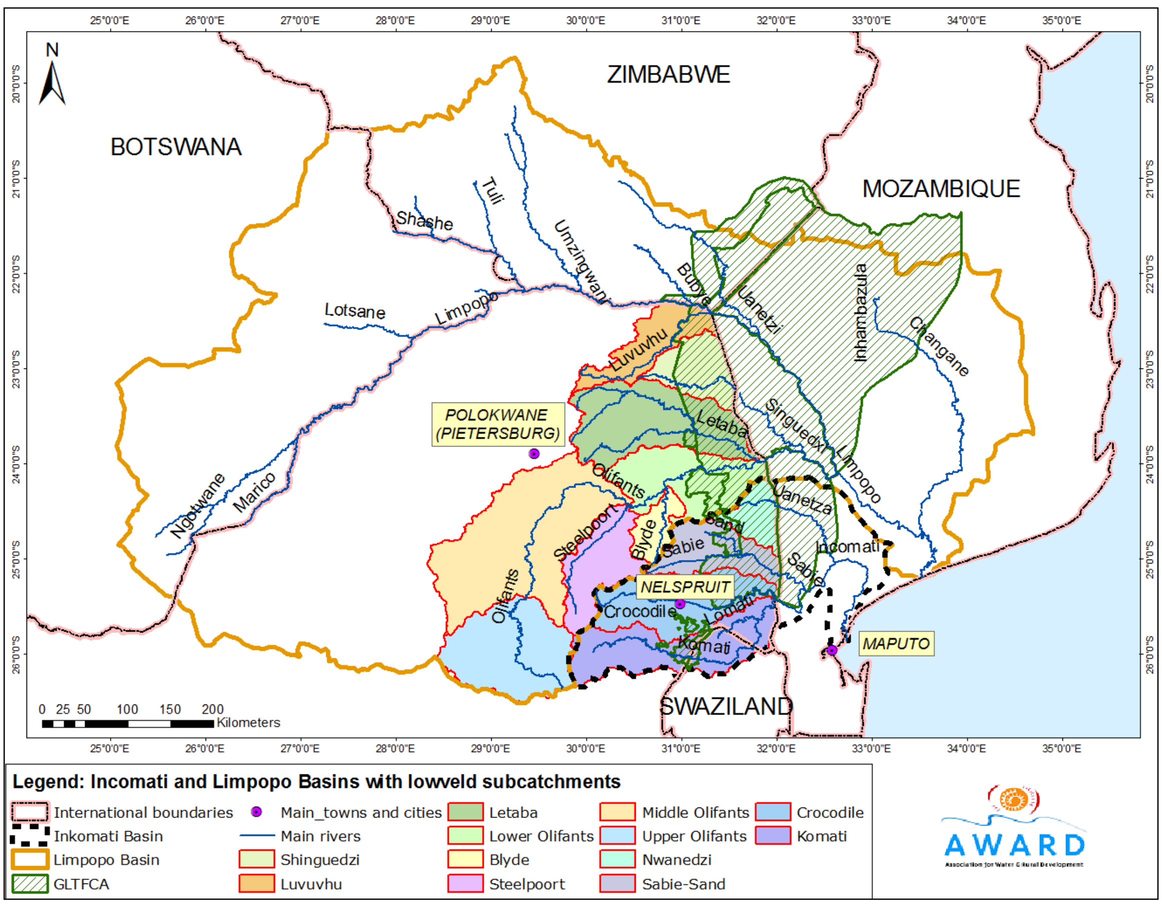


Figure 3 Six Lowveld rivers, WMAs and international basins (overview map)

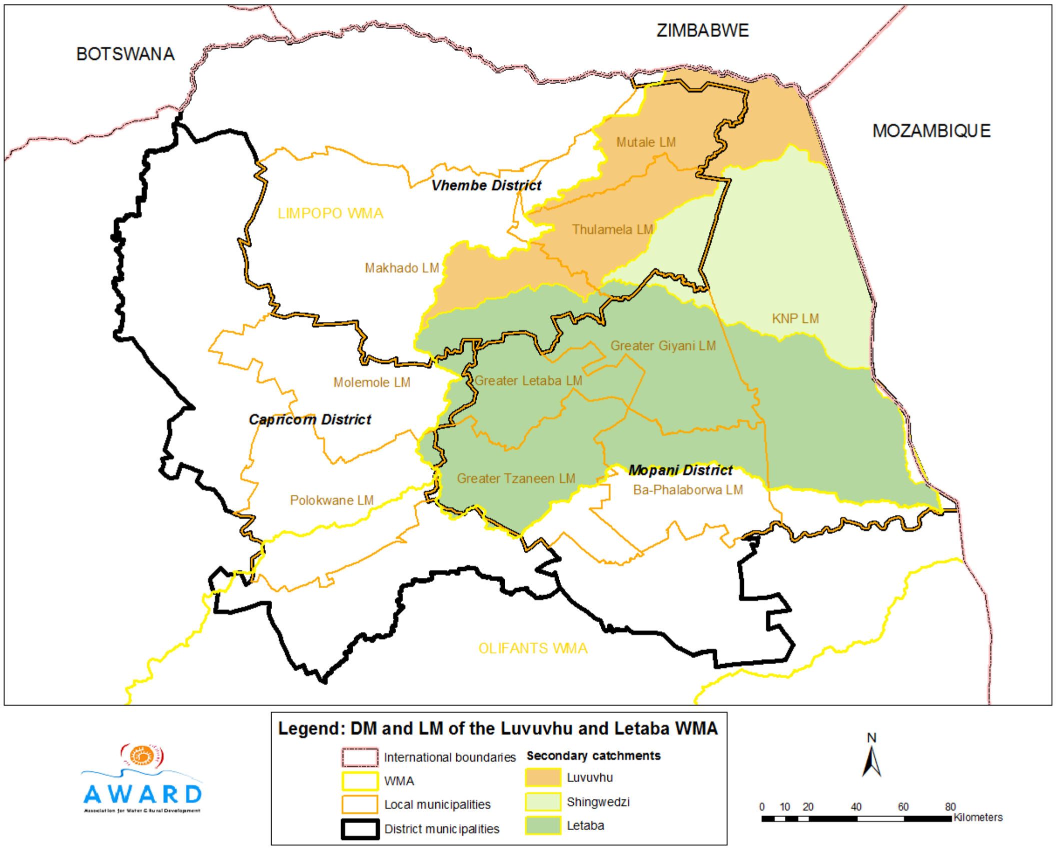


Figure 4 The non-alignment of northern rivers with district muicipality (administrative) boundaries

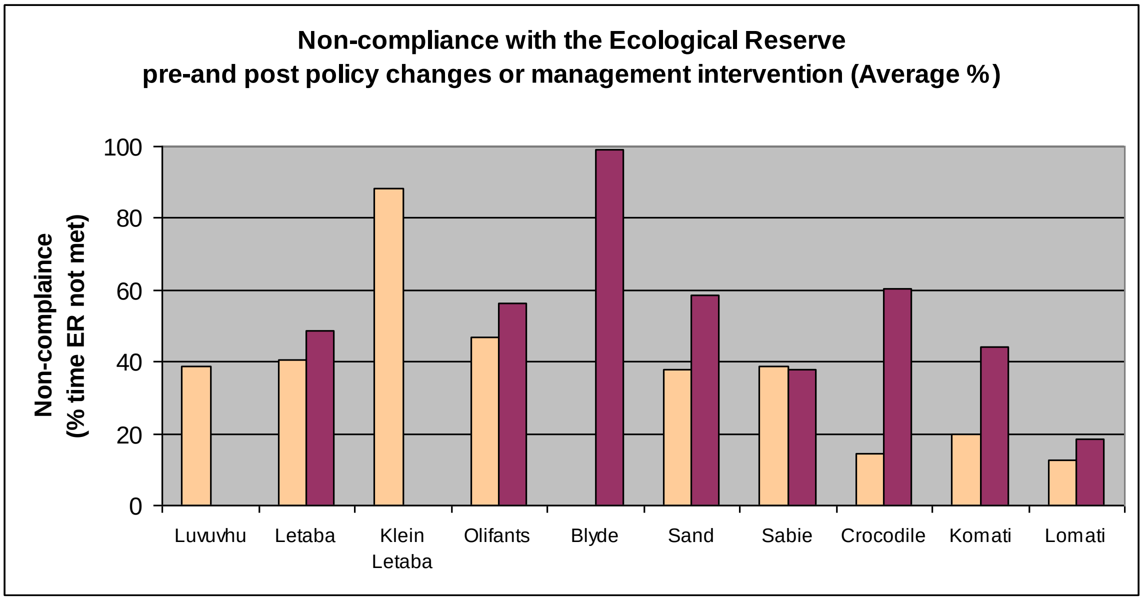


Figure 5 Compliance with the Ecological Reserve

Table 1 Visual comparison of issues in the catchments of the Lowvveld

Luvuvhu/ Letaba	Olifants	Inkomati	Transboundary
			
			

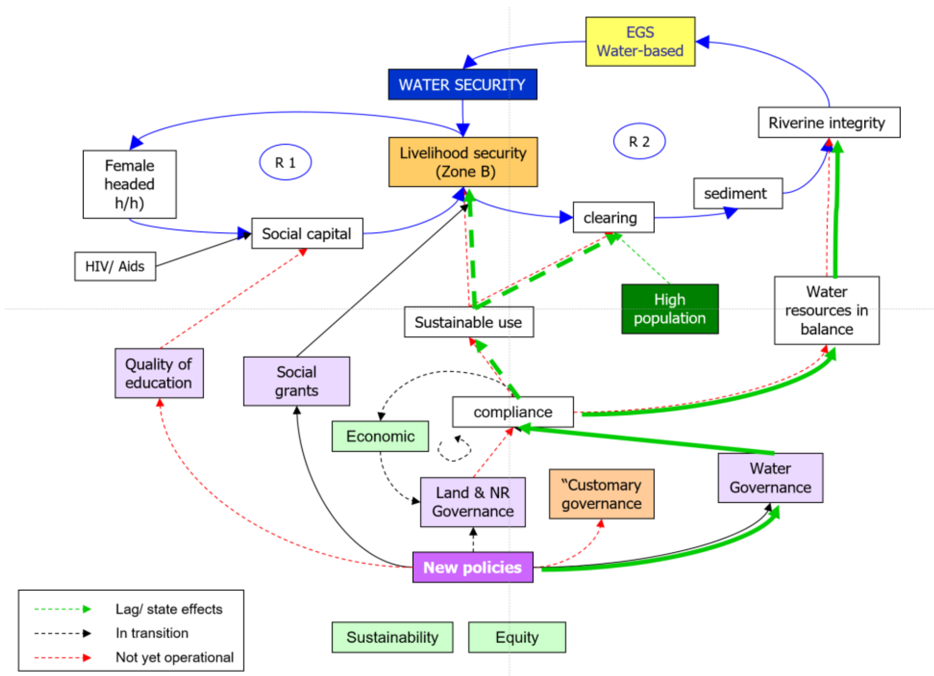


Figure 6 A typical complexity view of the catchments of the Lowveld

7. Catchment profiles

Each profile below is structured consistently: (1) a brief context summary, (2) DIKIW evidence showing how related datasets/experiences translate into understanding, and (3) a set of example Catchment Wisdom statements—guidance with clear adopters and conditions. The intention is not to provide a full technical assessment for each river, but to demonstrate a repeatable method for turning evidence into decisions and ‘catchment wisdom’.

a) Luvuvhu River (Limpopo/Olifants WMA)

Context

The Luvuvhu is a northern Lowveld river within the Limpopo/Olifants WMA, supporting irrigated agriculture, growing municipal demand, and downstream ecological systems connected to the greater Limpopo basin (Kruger Park and Mozambique). Key infrastructure—including Nandoni Dam—has increased the capacity to supply domestic and productive uses, but has also introduced a classic ‘allocation before ecology’ challenge: in the SRI evidence base, the dam was largely allocated prior to a robust incorporation of the Ecological Reserve into operating and allocation decisions. The catchment also contains significant opportunities for increasing available water through ecosystem restoration interventions such as alien invasive plant clearing.

Table 2Summary of the water resources, demand and balance from the ISP (DWAF, 2004c)

Availability/ use	Luvuvhu	Mutale	Groot Letaba	Klein Letaba	Lower Letaba
MAR	520	see Luvuvhu	382	151	42
Total local yield	**143**	**27**	**159**	**32**	
Use					
Urban	5	0	3	3	
Rural	9	2	10	8	
Industry/Mining	0	1	0	0	
Irrigation	83	24	133	25	
Afforestation	6	1	35	1	
Total demand	**103**	**28**	**181**	**37**	
Transfer in	4	5	0	0	0
Transfer out	7	4	15	0	0
Balance (incl. Desktop Reserve)	**37**	**0**	**-37**	**-5**	**0**

DIKIW evidence (Data → Information → Knowledge → Insight → Wisdom)

- Data: flow records at Luvuvhu EWR sites; Nandoni Dam storage and release records; abstraction and allocation data (irrigation and municipal); rainfall and seasonal climate outlooks; invasive alien plant extent and clearing records; stakeholder reports on supply interruptions and curtailments.
- Information: Reserve compliance summaries by month/season; water-balance and allocation summaries; maps of demand nodes and restoration opportunities (e.g., invasive alien plant areas); dashboards linking releases and downstream response; scenario outputs comparing allocation choices against Reserve and basic human needs.
- Knowledge: shared interpretation that compliance and reliability depend on the alignment of dam operations, allocations and reconciliation planning, and that 'new' water from infrastructure can be absorbed by demand unless explicit protections are built in.
- Insight: where infrastructure is fully allocated prior to Reserve incorporation, achieving compliance requires both operational optimisation and institutional/legal action (reconciliation, reallocation, and possibly compulsory licensing), alongside restoration measures that increase effective supply.
- Wisdom: explicit, accountable guidance that (a) prevents further 'allocation-before-Reserve', (b) integrates municipal water services planning with water resources constraints, and (c) treats restoration gains as part of a transparent water-accounting system.

Insights from from the Luvuvhu catchment

Reserve compliance (Reserve audit): the Luvuvhu showed an average incidence of failure to meet the Reserve of about 38.8%, with high failure in September (≈78.4%) and relatively low failure in December (≈9.1%). This indicates a strong seasonal compliance risk, particularly in late dry-season conditions.

Allocation and operating-rule gap: SRI notes that although the White Paper motivation for Nandoni Dam included recognition of environmental water needs, the intention was vague and the dam was largely allocated before the Reserve requirement was effectively embedded in operations and planning. SRI suggests that securing the Reserve could be technically possible through appropriate operation, but that existing allocations—including temporary allocations to municipal schemes such as Giyani—make this difficult without a decisive reconciliation process.

Need for integrated planning: SRI highlights that achieving Reserve and basic human needs outcomes in the Luvuvhu requires integrated planning across dam operations, water services planning, and the emerging CMA mandate. It flags that recalling allocations (where required) is unlikely without formal processes such as compulsory licensing.

Restoration leverage: SRI identifies substantial invasive alien plant coverage (≈168 km^2) and notes that clearing would increase resource availability—an important 'supply-side' complement to governance and demand management.

Constraints and risks

- Legacy allocations that constrain operational flexibility: once water is committed, Reserve protection becomes politically and legally complex.
- Institutional capacity and transition: as CMA functions emerge, there is risk of role ambiguity and delayed implementation.
- Demand growth (municipal and irrigation) outpacing reconciliation planning.
- Seasonal stress: late dry-season conditions amplify non-compliance risk and conflict potential.

Key areas for action

- **Use the Nandoni operations review as a flagship DIKIW process.** Combine technical modelling with stakeholder validation to convert data into agreed information, shared understanding, and practical guidance that can be applied in day-to-day operations and allocation decisions.

- **Pair allocation reconciliation with visible, early gains to build legitimacy.** Link tougher decisions on allocations and restrictions to quick, tangible improvements—especially invasive alien clearing and municipal demand management (WCDM/NRW reduction)—so stakeholders see progress and fairness.

- **Adopt minimum common monitoring products to reduce uncertainty and enable early correction.** Establish a small set of routine, shared outputs—such as a monthly compliance brief and a simple dashboard—so all parties work from the same evidence base and can respond before problems escalate.

- **Build legal and regulatory support into the catchment programme from the start.** Put verification, licensing clarity, and agreed enforcement/escalation protocols in place early so cooperation is backed by credible compliance and implementation does not stall on uncertainty or weak authority.

Wisdom statements for the Levuvhu

1. **Stop "allocation before ecology": explicit Reserve incorporation before new allocations from Nandoni-related systems.**
   SRI indicates that allocating storage before embedding the Reserve creates long-term compliance and legitimacy problems. This should be adopted by regulator/CMA licensing units, dam operators, municipalities, and irrigation bodies, and should only proceed with a reconciled water budget (including BHNR + Reserve), transparent allocation criteria, and routine compliance reporting.

2. **Use an integrated reconciliation package for municipal schemes and irrigation.** Technical operation alone cannot secure the Reserve where allocations already exceed what the system can supply under dry-season conditions. This should be adopted by CMA/DWS, municipalities/WSPs, local leadership, and user associations, and should include compulsory licensing readiness, metering/verification, WCDM and NRW reduction, and equity safeguards.

3. **Treat invasive alien plant clearing as “real water” with accountable water accounting.** Clearing can increase availability, but benefits must be measured and governed to prevent immediate re-absorption by new demand. This should be adopted by catchment restoration programmes, CMA planning, municipalities, and landholders, and requires baseline mapping, monitored yield gains, agreed benefit-sharing, and protection of the Reserve and BHNR as first calls.

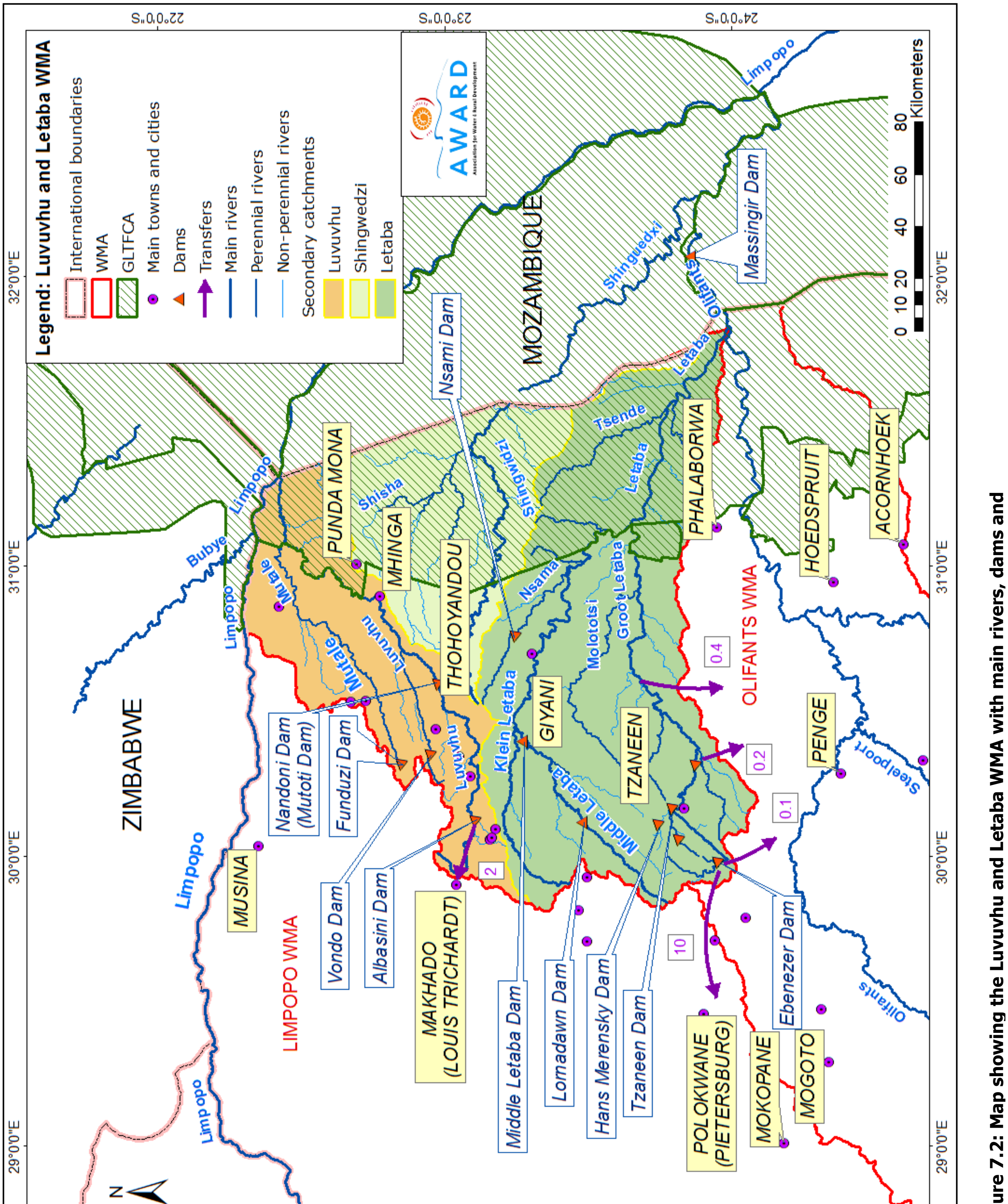


Figure 7 Map of the northern catchments of the Lowveld

b) Letaba River (Limpopo/Olifants WMA)

Context

The Letaba is a water-stressed Lowveld river system with major irrigation nodes, rural and peri- urban domestic supply demands, and high downstream ecological importance. It includes distinct sub-systems with very different stress profiles, notably the Groot Letaba (with improving management and partial compliance gains) and the Klein/Middle Letaba, which the SRI work describes as being in near-crisis under chronic deficits. Key infrastructure (including Tzaneen Dam and the Middle Letaba Dam) shapes release patterns, while institutional arrangements span water resources management, water user associations, and municipal water services planning.

DIKIW evidence (Data → Information → Knowledge → Insight → Wisdom)

- Data: daily flows at Letaba EWR sites (including downstream reaches near the western boundary of Kruger National Park); storage and release records for Tzaneen Dam and the Middle Letaba Dam; abstraction volumes (irrigation and municipal); rainfall and seasonal outlooks; restriction notices; forum minutes and stakeholder narratives about shortages and curtailments.
- Information: Reserve compliance summaries by sub-system (Groot vs Klein/Middle Letaba); time- series plots of flow shortfalls and dry- season failure; operational comparisons of ‘minimum flow rules’ vs determined Reserve requirements; water-balance summaries including transfers and allocations.
- Knowledge: shared interpretation that compliance outcomes depend on both physical constraints (chronic deficits, lags between dam release and downstream response) and institutional practice (how rules are interpreted, communicated and enforced).
- Insight: static ‘minimum flow’ rules can produce a false sense of compliance; dynamic, rainfall- responsive Reserve requirements and lag-aware operations are required, and must be paired with cooperative governance and credible compliance measures—especially in the Klein/Middle Letaba.
- Wisdom: an integrated operating-and-allocation approach that (a) updates rules to reflect dynamic Reserve requirements, (b) aligns municipal water-supply planning with water-resources constraints, and (c) builds legitimacy for restrictions through transparent triggers and shared forums.

Key examples from the Shared Rivers Initiative (SRI)

Reserve compliance snapshot (Reserve audit): Groot Letaba—average incidence of failure ≈24.6% (≈53.6% in September and ~0% in December), with SRI noting a steady improvement over time linked to improved management and stakeholder buy- in. Klein Letaba—average incidence of failure ≈88.4% (≈91.6% in September and ≈78.1% in December), reflecting persistent infringement and severe deficits.

Static vs dynamic Reserve interpretation: SRI reports that operational management in parts of the Letaba was effectively based on a constant minimum flow value (reported as ≈0.6 m³/s), whereas the determined Reserve requirements are dynamic and rainfall-responsive. This gap between operational practice and Reserve determination contributed to recurring shortfalls.

Lag and responsiveness: SRI highlights the operational challenge that releases from Tzaneen Dam can take about seven days to reach downstream reaches, limiting the ability to respond quickly to observed shortfalls near key ecological nodes. This motivated work toward a more responsive, system-based operations approach.

Klein/Middle Letaba near-crisis: SRI notes that the Middle Letaba Dam can be at ~10% or less, and that in the Klein/Middle Letaba the Reserve was met less than 10% of the time in the dry season in some analyses, with little evidence of effective mitigation under increasing demand pressure.

Key constraints and risks

- Chronic deficits and high competition: even small demand increases can translate into large ecological and downstream impacts, especially in the Klein/Middle Letaba.
- Political sensitivity of restrictions and reallocations: curtailment decisions can be contested unless the rationale and process are shared and equitable.
- Institutional fragmentation: water resources operations, water user associations and municipal planning can work in parallel without a single reconciled plan.
- Enforcement gaps and unlawful use: where verification, metering and consequences are weak, voluntary compliance is insufficient in high- stress periods.

Enablers and leverage points

- Strengthen and professionalise catchment forums as action-oriented learning spaces (not only consultation platforms).
- Use compliance evidence to negotiate 'variable assurance of supply' and cooperative curtailment agreements before drought peaks.
- Target municipal demand management and non- revenue water reduction as a high-leverage pathway (especially where basic human needs are threatened).
- Translate Reserve requirements into a small set of operational targets and triggers that reflect lag times and can be tracked publicly.

Example Catchment Wisdom statements (adoptable guidance)

• 1) Replace static minimum-flow practice with dynamic, lag-aware Reserve operations.

- Because: Because SRI evidence shows that constant minimum-flow rules can diverge from the determined Reserve, and lag times make reactive management ineffective.

- For whom: Primary adopters: dam operators; DWS/CMA operations teams; Letaba forum/WUA leadership.

- Conditions for success: Conditions: translation of dynamic Reserve into operational targets; pre-dry-season release planning; monitoring aligned to EWR sites; review after each dry season.

• 2) Establish an integrated Letaba 'operations + allocation + water services' reconciliation platform.

- Because: Because municipal planning and water-resources management must be aligned to avoid demand growth absorbing ecological water and to protect basic human needs.

- For whom: Primary adopters: CMA/DWS; municipalities and WSPs; WUAs; conservation agencies; catchment forums.

- Conditions for success: Conditions: shared water-balance and demand projections; agreed priorities (BHN + Reserve); transparent restrictions; joint communication.

• 3) In the Klein/Middle Letaba, prioritise compulsory licensing / verification and a credible compliance package.

- Because: Because SRI describes persistent, severe infringement where voluntary measures are insufficient under chronic deficit.

- For whom: Primary adopters: regulator/CMA compliance units; WUAs; local leadership structures.

- Conditions for success: Conditions: metering/telemetry; clear enforcement pathway; equity safeguards; support for affected users to adapt (efficiency, crop changes, alternative supply).

c) Olifants River (Limpopo/Olifants WMA)

Context

The Olifants is a large, highly developed Lowveld river with intensive irrigation, major mining and industrial activity, and significant power-sector water use. It is also transboundary, flowing into Mozambique and supporting downstream ecosystems and livelihoods. Within the SRI evidence base, the Olifants stands out for the tight coupling of flow management with water quality risks (acid mine drainage, wastewater treatment failures, and mining-related contamination), making it a flagship case for 'quantity–quality–governance' integration.

The Olifants catchment has a surface area of approximately 54 550 $km^{2.}$ It comprises seven secondary catchments (B4) which show significant variations in climate, water availability, level and nature of economic development, and population density. As such DWS (DWAF, 2004d) divided the WMA into sub-areas to facilitate improved management of water resources. We have further separated the Blyde sub-area from the Lower Olifants as this has unique characteristics. Thus this report will discuss the five sub-catchments, namely the Upper Olifants, Middle Olifants, Steelpoort, Blyde and Lower Olifants (Figure 3.5 and 3.6).

- The **Upper Olifants** which constitutes the catchment of the Olifants River from the source down to Loskop Dam.
- The **Middle Olifants** comprises the area downstream of Loskop Dam to the confluence of the Steelpoort River, a (river) distance of approximately 302 km.
- The **Steelpoort** Catchment
- The **Blyde** Catchment bounded by the Steelpoort and Lower Olifants boundaries.
- The **Lower Olifants** represents the catchment of the Olifants River between the foothills of the Drakensberg Escarpment and the Mozambique border.

Table 3 Summary of the water resources, demand and balance for the Olifants WMA (DWAF, 2004d)

Availability/ Use	**Upper Olifants**	**Middle Olifants**	**Steelpoort**	**Lower Olifants**	**Total**
Total local yield	238	210	61	100	**609**
Transfers in	171	92	0	1	264
Grand Total Water Availability	**409**	**302**	**61**	**101**	**873**
Use					
Irrigation	44	336	69	108	**557**
Urban	62	15	3	7	**87**
Rural	6	28	6	5	**45**
Mining and industrial	20	13	17	43	**93**
Power Generation	181	0	0	0	**181**
Afforestation	1	0	1	1	**3**
Total requirements	**314**	**392**	**96**	**164**	**966**
Transfers out	96	3	0	0	**8**
Grand Total	**410**	**395**	**96**	**164**	**974**
Balance	**-1**	**-93**	**-35**	**-63**	**-192**

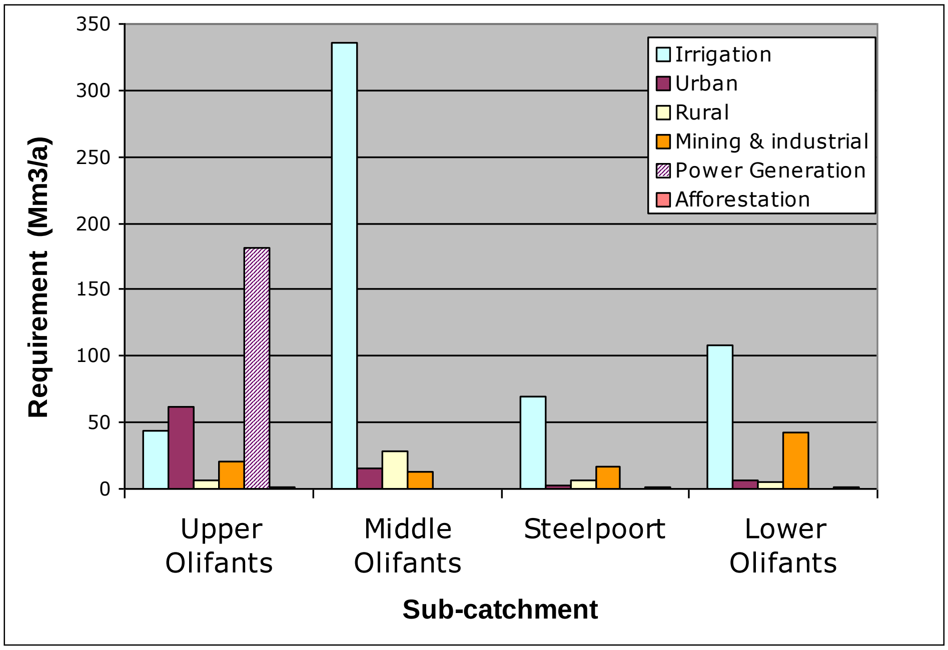


Figure 9 Water requirements in the Olifants WMA (2000)

Figure 10 Map of the Olifants - the largest of the catchments

DIKIW evidence (Data → Information → Knowledge → Insight → Wisdom)

- Data: flow records at key EWR sites (including lower Olifants); dam storage/release data (e.g., Loskop and other system dams); abstraction volumes by sector; water quality measurements (e.g., salinity, metals, nutrients); mine closure/decant estimates; incident reports (fish kills, crocodile deaths, pollution events).
- Information: sectoral water-use breakdowns; Reserve compliance summaries (including zero/near-zero flow days); trend analyses of compliance changes across periods; water quality trend charts; risk maps for pollution hotspots and vulnerable reaches.
- Knowledge: shared interpretation that Reserve shortfalls and water quality deterioration are jointly driven by high demand, operational rule gaps, and insufficiently integrated regulation across water resources and pollution control domains.
- Insight: in the Olifants, 'meeting the Reserve' is not only an operations problem; it requires coordinated governance across mining regulation, municipal wastewater management, dam operations, and abstraction control—otherwise ecological impacts persist even when some flow targets are met.
- Wisdom: integrated, enforceable action packages that link operating rules, demand controls and pollution controls, with clear roles, escalation steps and transparent reporting across sectors and borders.

Key examples from the Shared Rivers Initiative (SRI)

Reserve compliance snapshot (Reserve audit): the lower Olifants showed an average incidence of failure to meet the Reserve of about 50% (with higher failure in September). SRI records a major low- flow incident where the lower Olifants ceased flowing for approximately 33 days during September–October 2005, illustrating the severity of dry- season risk when operations and abstractions are not aligned to downstream requirements.

Sectoral pressure: SRI reports that the Olifants' registered water use is dominated by irrigation (≈48%), mining (≈25%) and power generation (≈20%), underscoring the need for multi-sector governance rather than single- sector fixes.

Quality risks as a governance amplifier: SRI highlights worsening acid mine drainage risk due to increased mining, combined with severe sewage impacts from poorly functioning wastewater treatment works. It also points to downstream ecological consequences, including major crocodile deaths in Mozambique following a period of contamination after the raising of the Massingir Dam wall. A projected mine- closure decant volume (≈62 Mm^3/a) is flagged as a future risk that would further increase pressure on the system.

Key constraints and risks

- Complex multi-sector politics: high economic stakes (mining, power generation, commercial agriculture) create resistance to restrictions and enforcement.
- Institutional fragmentation across quantity and quality mandates: flow management and pollution regulation often operate in parallel rather than as one risk system.

- Legacy infrastructure and allocation patterns that pre-date robust Reserve implementation.
- Transboundary impacts: downstream consequences can amplify reputational and diplomatic risk when compliance and water quality deteriorate.

Enablers and leverage points

- Use compliance evidence (including cessation-of-flow events) as a 'shared reality' that motivates coordinated action.
- Integrate operational planning with pollution-control priorities: align release decisions with dilution/assimilation needs and wastewater risk windows.
- Target high-leverage nodes: prioritise upgrades at the worst-performing wastewater treatment works and the highest-risk mining points, alongside abstraction control.
- Create a cross-sector, cross-border learning-and-response platform that routinely synthesises evidence into action commitments.

Example Catchment Wisdom statements

1) Manage water quality and ecological flows as one compliance package.

- Because SRI evidence shows that pollution risks (AMD and sewage) can undermine ecological outcomes and trigger downstream crises even when some quantity targets are met.
- Primary adopters: CMA and DWS operations; environmental regulators; municipalities; mining regulators and major operators.
- Conditions: joint risk dashboard; coordinated enforcement; investment plans tied to compliance; incident response protocols.

2) Establish dam operating rules that translate Reserve requirements into specific, monitored release targets at multiple nodes.

- Because recurring shortfalls and cessation-of-flow events indicate that existing operating practice is not consistently protecting downstream needs.
- Primary adopters: dam operators; CMA operations teams; sectoral users participating in restriction planning.
- Conditions: scenario planning for dry seasons; clear triggers and escalation; routine publication of outcomes vs targets.

3) Prioritise 'hotspot enforcement' and verification where pressure is highest.

- Because demand is concentrated in a few large sectors and locations, and targeted compliance interventions can yield system-level benefits.
- Primary adopters: regulator, CMA compliance units, sectoral associations.
- Conditions: reliable metering/telemetry; legal support and clarity; transparent penalties and incentives; protection of basic human needs and the Reserve.

d) Sabie-Sand system (Inkomati/Usutu WMA)

Context

The Sabie Sand is part of the Inkomati/Usustu WMA (Figure 3.9). It is divided into four sub-catchments[1], as follows.

1. The **Sabie-Sand River Catchment** which lies in the north of the WMA. The Sand River is the main tributary of the Sabie.
2. The **Crocodile Catchment** (includes the regional capital of Nelspruit).
3. The **Komati Catchment**. The Komati River, which rises in South Africa, flows through Swaziland and then re-enters South Africa before flowing on into Mozambique;
4. The **Usustu Catchment** lies in part of Notrthern KwaZulu Natal. The Usutu was added later and data is not included in this water balance study.

The confluence of the Crocodile and Komati rivers lies just upstream of the Mozambique border at Ressano Garcia where after the river is known as the Incomati River. The Sabie flows into the Incomati below Corumana Dam near the town of Moamba.

Table 4 Water availability and demand and water balance of the Inkomati WMA including the Reserve estimates (based on the preliminary estimate 2008 (DWAF, 2009; IWAAS, 2009). SFRA = Stream flow reduction activity

Availability/ Use	X1: Komati	X2: Crocodile	X3: Sabie	Inkomati WMA
Availability	775	555	116	1446
Current use (excl Reserve)	**858**	**632.3**	**179.5**	**1670**
Allocated use with Reserve				
Cross Border	62	51	0	112
Reserve	228	205	209	
Domestic	47	73	82	202
Industry/Mining	2	27	0	29
Irrigation	642	482	98	1222
Strategic	105	0	0	105
Total demand with Reserve	**1086**	**837**	**389**	**2311**
Afforestation	117	158	90	365
Alien Vegetation	32	32	16	80
Balance currently	**-83**	**-77.3**	**-63.5**	**-223.8**
Balance with Reserve	**-311**	**-282**	**-273**	**-865**

[1] The undeveloped Nwanedzi River that is wholly within the Kruger National Park and is not considered in this report.

Figure 11 : Map showing the Inkomati WMA with main rivers, dams, water transfers and the Inkomati Basin.

The Sabie Sand subcatchments

The Sabie–Sand system is internationally recognised for high ecological and tourism value, with critical downstream assets in/adjacent to protected areas. Within the broader system, the Sabie River has generally higher flows and better overall compliance than most Lowveld rivers, while the Sand River is a much more stressed, demand-dominated sub-catchment. The SRI work treats this pairing as a powerful comparative case: what 'good' looks like in a relatively better-performing river (Sabie) and what chronic dry-season stress looks like where demand and governance arrangements differ (Sand).

DIKIW evidence (Data → Information → Knowledge → Insight → Wisdom)

- Data: daily flow records at Sabie and Sand EWR sites; releases and storage for key infrastructure (including Injaka Dam and related systems); municipal offtake volumes; irrigation abstractions; rainfall and seasonal outlooks; stakeholder meeting records from local forums and operating meetings.
- Information: compliance summaries by month and by river (Sabie vs Sand); plots of near-zero days and dry-season deficits; water-balance snapshots; dashboards linking augmentation releases to downstream response; maps showing offtake nodes and monitoring points.
- Knowledge: shared understanding that (a) the Sabie is an exception but still vulnerable in dry months, and (b) the Sand shows persistent dry-season failure driven by high local demand and weakly integrated operational controls.
- Insight: infrastructure and 'augmentation intentions' do not guarantee ecological outcomes; without coordinated water-resources management, water-supply planning, and enforceable operating rules, freed-up water is rapidly re-absorbed by demand and does not 'travel' to the river.
- Wisdom: practical operating integrity and governance measures—simple, observable and enforceable—that protect dry-season flows and prevent 'paper compliance' (plans and rules without outcomes).

Key examples from the Shared Rivers Initiative (SRI)

Reserve compliance snapshot (Reserve audit): the Sabie showed a low average incidence of failure (≈6% overall), with failures concentrated in dry months (≈14.8% in September and ~0% in December). By contrast, the Sand showed persistent failure (≈61% overall), with very high dry- season failure (≈80.5% in September and ≈42.6% in December). SRI treats the Sand as a distinct case because of recognised biophysical, social and institutional differences.

Operating-rule and 'intention-to-outcome' gaps: SRI notes that the original intentions for Injaka (augmentation for domestic supply and ecological support) were not consistently realised because operating rules were not fully implemented or adhered to, and because increasing urban/municipal demand and weak coordination reduced the water that actually travelled downstream.

Monitoring and reconciliation challenges: SRI highlights the difficulty of assessing compliance when gauging stations are far downstream of EWR sites and when there are net losses and multiple offtakes between dam and gauge. This complicates both operational decision-making and accountability.

Key constraints and risks

- Demand creep: as domestic and municipal demand grows, ecological and downstream needs can be squeezed unless explicitly protected.
- Weak integration between water-resources operations and water-supply planning: infrastructure upgrades (and water savings) can be rapidly absorbed by new demand if not governed.
- Measurement blind spots: long distances between operational control points, EWR sites and gauges can mask deficits and delay corrective action.
- Fragmented governance: multiple institutions and programmes can act in parallel without coherent reconciliation and prioritisation.

Enablers and leverage points

- High shared value of ecological outcomes (tourism, ecosystem services, cultural value) can provide a unifying narrative for cooperation.
- Use the Sabie as a 'reference case' and the Sand as a 'stress-test' to design governance measures that work under scarcity.
- Strengthen operating-rule adherence through simple 'integrity checks' (targets vs observed at key points; actions taken; reasons).
- Improve monitoring design: align gauges and reporting to EWR sites and key offtake nodes; include community observation as supplementary evidence.

Example Catchment Wisdom statements (adoptable guidance)

1) Create an 'operating integrity protocol' for Injaka augmentation and dry-season flows.

- Because: Because SRI evidence shows that augmentation intentions can be lost without operating-rule adherence and integrated demand control.

- For whom: Primary adopters: Inkomati CMA operations; dam operators; municipalities; local forums.

- Conditions for success: Conditions: explicit dry- season targets at agreed sites; routine reporting of releases/abstractions; corrective actions logged; annual review.

2) In the Sand, link municipal expansion and infrastructure upgrades to Reserve compliance tests.

- Because: Because water freed through upgrades is quickly taken up unless the reconciliation process explicitly protects ecological and downstream outcomes.

- For whom: Primary adopters: municipalities and WSPs; CMA licensing and planning teams; provincial/national water services planning units.

- Conditions for success: Conditions: require 'no- harm' compliance evidence for new offtakes; integrate WCDM and non- revenue water reduction; transparent curtailment triggers.

3) Redesign monitoring to be decision-ready (EWR- aligned), not only report-ready.

- Because: Because compliance assessment and operational control are undermined when gauges do not reflect what is happening at EWR sites.

- For whom: Primary adopters: hydrology services, CMA monitoring teams, research partners.

- Conditions for success: Conditions: agreed monitoring design; metadata; rapid data access; community observation integration where appropriate.

e) Crocodile River (Inkomati/Usutu WMA)

Context

The Crocodile River is one of the most economically productive and institutionally complex rivers in the Lowveld, supporting intensive irrigation, fast-growing urban demand (including Mbombela/Nelspruit), and downstream ecological assets linked to the Inkomati system. Its flow regime is heavily shaped by storage (notably Kwena Dam), large-scale abstractions, and operating decisions that must balance short-term supply reliability with the Ecological Reserve. The SRI work treats the Crocodile as a critical test-case for turning Reserve determinations into day-to-day operational practice.

The greatest demand for water is from irrigated agriculture and forestry. In terms of water infrastructure the catchment has one major dam, the Kwena Dam, in the upper catchment (which augments low flows) and a number of smaller dams in the central portion (Witklip, Primkop, Klipkoppie/ Longmere;).

The water requirements exceed the available resource, and the catchment is considered to be **highly stressed**. The IWAAS conclusion is that the irrigation demands have been increasing since the 1990's up to their current levels. DWA's current policy for many years has been not to issue any more water use licences to irrigation but there is probably still some unlawful development.

Currently there is a real time study underway to address key problems east of the Kwena Dam. The objectives of this study – known as the *Real Time Operating Decision Support System for the Crocodile East River System* – are to assist with water distribution (run of river) and water releases (dams), to ensure compliance with the Reserve and with international obligations (Crocodile East RTOS meeting Nov 2007). The DSS must be capable of determining operational plans and should include a water allocation and utilisation management and monitoring system (DSS Team pers. comm.).

DIKIW evidence (Data → Information → Knowledge → Insight → Wisdom)

- Data: daily flow records at Crocodile EWR sites (e.g., Tenbosch); dam storage and release data (Kwena); abstraction records across irrigation and urban nodes; electricity tariff schedules and pumping patterns; restriction notices and compliance logs; minutes from the Crocodile River Forum and related platforms.
- Information: monthly/seasonal Reserve compliance summaries; trend plots showing increases in failure incidence; dashboards linking releases and abstractions to downstream targets; analyses of intra-day variability associated with pumping; simple risk indicators for upcoming dry-season stress.
- Knowledge: agreed interpretation that persistent non-compliance is driven by overallocation (historic allocations and trading), increasing demand (irrigation and urban), and operational practice that is not yet consistently aligned with Reserve targets.
- Insight: technical tools (decision support systems) only work when paired with cooperative governance, transparent reporting, and enforceable curtailment protocols; incentives (e.g., off- peak pumping) can create ecological impacts even when total volumes are within expectations.
- Wisdom: a resourced, reviewable operating and restriction regime—backed by forums and compliance tools—that translates Reserve requirements into decisions, obligations and consequences.

Key examples from the Shared Rivers Initiative (SRI)

Reserve compliance snapshot (Reserve audit): average incidence of failure to meet the Reserve in the Crocodile was reported at about 64.6%, with very high failure in September (≈87.9%) and much lower failure in December (≈15.6%). The SRI analysis indicates that failure incidence increased markedly across successive periods (e.g., 1994–2000 and 2001–2008), reflecting increasing pressure and the limits of existing operating practice.

Drivers behind non-compliance: SRI identifies multiple interacting drivers—rapid growth in irrigation and urban demand, a legacy of overallocation (including trading), and operational patterns that can be shaped by energy pricing and off- peak pumping. These drivers combine to produce sustained dry- season stress and recurring shortfall at downstream EWR sites.

Emergent solution pathway: the SRI work highlights the move toward near real- time decision support systems to improve adherence to operating rules and compliance with the Reserve. Importantly, this is framed as both a technical and institutional shift: decision support is most effective when it is embedded in cooperative governance and backed by monitoring and enforcement.

Key constraints and risks

- Legacy overallocation and contested legitimacy of reallocation: restrictions can be politically sensitive and can trigger conflict unless the rationale is shared and the process is seen as fair.
- High transaction costs of coordination: multiple sectors and institutions mean that slow decision-making can compound deficits during rapidly changing conditions.
- Intra-day variability: pumping incentives can undermine ecological objectives even when monthly volumes are managed, and can be difficult to detect without appropriate monitoring.
- Enforcement capacity: without reliable metering/telemetry and clear escalation procedures, operating rules can remain aspirational.

Enablers and leverage points

- Existing platforms (e.g., Crocodile River Forum / Lowveld forums) that can be strengthened as learning-and-action infrastructure.
- Near real-time ODSS / DSS capability as a backbone for transparent decision-making (linking releases, abstractions and downstream targets).
- A ‘minimum common product’ approach: monthly compliance brief + shared dashboard + seasonal restriction plan, enabling rapid alignment without excessive analysis.
- Legal and compliance tooling (verification, metering, and—where necessary—water bailiff support) to move from voluntary adherence to accountable practice.

Example Catchment Wisdom statements (adoptable guidance)

• 1) Make the ODSS/DSS the shared ‘single source of truth’ for restrictions and releases.

- Because: Because high-stress systems need fast, transparent decisions and a common evidence base; SRI highlights DSS as central to improving Reserve compliance.

- For whom: Primary adopters: Inkomati CMA operations; DWS dam operators; municipalities; irrigation bodies; Crocodile River Forum secretariat.

- Conditions for success: Conditions: agreed data-sharing; routine publication of key indicators; forum-backed decision rules; periodic audits of model assumptions and outcomes.

• 2) Treat pumping incentives and intra-day variability as a managed risk.

- Because: Because flow variability driven by tariff-linked pumping can create ecological stress and erode compliance credibility.

- For whom: Primary adopters: irrigator associations; energy utility counterparts; CMA operations; regulator.

- Conditions for success: Conditions: monitoring that captures variability; negotiated revisions to pumping agreements; triggers for corrective action.

• 3) Use 'variable assurance of supply' thinking to build cooperative curtailment agreements.

- Because: Because drought and low- flow management works better when users understand that access varies with availability, and when curtailment rules are agreed in advance.

- For whom: Primary adopters: CMA, irrigators, municipalities, conservation agencies, forum participants.

- Conditions for success: Conditions: seasonal forecasting/early warning; transparent curtailment triggers; clear communication; a credible enforcement backstop.

• 4) Institutionalise a compliance-and-learning loop through the Crocodile River Forum.

- Because: Because sustained improvement depends on repeated reflection, adjustment of rules, and collective legitimacy for hard decisions.

- For whom: Primary adopters: Crocodile River Forum leadership; CMA; local government and user groups.

- Conditions for success: Conditions: resourcing for convening and synthesis; structured reflection sessions; annual review of operating rules and restriction outcomes.

f) Komati River (Inkomati WMA, transboundary focus)

Context

The Komati is a transboundary Lowveld river in the Inkomati system, with downstream obligations to Mozambique and upstream interactions with Eswatini/Swaziland. Its flow regime is strongly shaped by multi-country infrastructure and operating arrangements (including Maguga and Driekoppies dams and associated offtake systems), and by irrigation and power-sector pumping incentives. In the SRI analysis, the key operational question is whether day-to-day dam and abstraction decisions are consistently aligned with the Ecological Reserve and with cross-border flow commitments.

The greatest water user in the Komati Catchment is irrigated agriculture (see Table 3.4). Pollard et al. 2010 provide figures that indicate the increase in irrigated agriculture in the Komati River particularly over the last two decades. This is accompanied by an increased demand for water. This is followed by stream flow reduction (forestry at 12%) and unique to the Komati, water for strategic use i.e. Eskom (10%). The two major dams Vygeboom and Nooitgedacht dams were built to provide water to the Eskom power Stations on the highveld. Most of the water from the upper Komati is for the use of Eskom. Alien vegetation has a significant impact on the water resources (3% if demand).

The ISP notes that the key issue in this sub-area is the transfer of water out of the WMA to the Olifants WMA. The NWRS reserves this transfer (i.e. requires national authorisation) up to **132 M m^3/a**, even though the current transfer is only about 97 M m^3/a. The implication is that transfers out of this sub-area could increase in future.

An important factor in the management of this sub-area is the position of **Swaziland downstream** of the sub-area. There is a treaty between South Africa and Swaziland, as well as the more recent Interim IncoMaputo Water Use Agreement, both of which influence the management of the water resources of this sub-area (see Chapter 5).

The lower Komati is considered to be highly stressed. However, the completion of the Maguga Dam brings the situation back into approximate balance. Implementation of the Reserve in the Komati catchment can be achieved now that the Driekoppies and Maguga dams are complete without resulting in large deficits in the lower catchment, but this is based on the assumption that the Reserve will also be implemented in the upper reaches, where much of the water for the Ecological Reserve originates.

Table 5 Water availability and demand and water balance of the Inkomati WMA including the Reserve estimates (based on the preliminary estimate 2008 (DWAF, 2009; IWAAS, 2009). SFRA = Stream flow reduction activity

Availability/ Use	X1: Komati	X2: Crocodile	X3: Sabie	Inkomati WMA
Availability	775	555	116	1446
Current use (excl Reserve)	**858**	**632.3**	**179.5**	**1670**
Allocated use with Reserve				
Cross Border	62	51	0	112
Reserve	228	205	209	
Domestic	47	73	82	202
Industry/Mining	2	27	0	29
Irrigation	642	482	98	1222
Strategic	105	0	0	105
Total demand with Reserve	**1086**	**837**	**389**	**2311**
Afforestation	117	158	90	365
Alien Vegetation	32	32	16	80
Balance currently	**-83**	**-77.3**	**-63.5**	**-223.8**
Balance with Reserve	**-311**	**-282**	**-273**	**-865**

DIKIW evidence (Data → Information → Knowledge → Insight → Wisdom)

- Data: daily flows at Komati EWR sites; dam storage and release records (Maguga/Driekoppies and related systems); abstraction volumes (irrigation, municipal, industry); rainfall and temperature; incident logs for near-zero flow events; minutes from transboundary operating meetings.
- Information: Reserve compliance summaries by month/season; period-comparison plots showing shifts in non-compliance; hydrographs of zero/near-zero days; dashboards linking releases/abstractions to downstream targets; simplified water-accounting summaries for the dry season.
- Knowledge: validated explanation of where the operating rule set does and does not embed Reserve requirements (including the difference between an operational minimum flow rule and a formally determined Reserve requirement), and when enforcement vs. rule revision is the binding constraint.
- Insight: transboundary compliance improves when rules are simple, jointly visible, and paired with routine, trusted reporting; where incentives (e.g., pumping schedules) create unintended variability, technical fixes must be coupled to negotiated governance adjustments.
- Wisdom: jointly agreed, reviewable operating guidance (e.g., updated rule curves and release targets; escalation steps when targets are missed; and minimum common reporting products that build trust without overburdening institutions).

Key examples from the Shared Rivers Initiative (SRI)

Reserve compliance snapshot (Reserve audit): average incidence of failure to meet the Reserve was reported at about 11.1% for the Komati (with higher failure in September and very low failure in December), but the SRI trend analysis highlights a shift: from failures largely concentrated in the wet season before ~1990, to failures occurring in all months thereafter, with much higher average failure rates in 2001–2008.

Operating rule gap: SRI notes that Reserve requirements were not fully incorporated into the operating rule sets being used at the time, which instead relied on an operational minimum flow requirement (reported as approximately 1.1 m^3/s). This mismatch between 'rules-in-use' and 'Reserve-in-law' is a core driver of recurring non-compliance.

Flow variability and incentives: the SRI work highlights that pumping incentives (including electricity tariff structures) can introduce intra-day and intra-week variability, creating ecological stress and complicating compliance even when average volumes appear adequate. This reinforces the need for rule revision plus transparent reporting.

Key constraints and risks

- Transboundary coordination burden: multiple operators and mandates can slow response during drought and create ambiguity about who adjusts releases first.
- Data and transparency gaps: where abstraction or release data are not routinely shared, trust erodes and compliance disputes become harder to resolve.
- Rule complexity: overly complex or poorly communicated rule sets reduce adherence and increase the likelihood of unintended non-compliance.
- Incentive misalignment: pumping schedules and sectoral optimisation (e.g., energy costs) can undermine ecological objectives unless explicitly governed.

Enablers and leverage points

- Use the transboundary setting as a forcing function for routine, standardised reporting (a simple monthly compliance brief and a shared dashboard).
- Rule review windows: schedule an annual (or pre-dry-season) rule review where Reserve requirements are explicitly translated into operational targets.
- Minimum common indicators across borders (Reserve compliance at agreed sites; zero/near-zero days; rule adherence checks).
- Create a clear escalation pathway (technical adjustment → negotiated curtailment → regulatory action) that is agreed in advance.

Figure 12 Maguga Dam in Swaziland along with the Driekoppies scheme provides an opportunity to manage the Komati with a higher level of assurance. The allocation of environmental flows within this system needs to be a priority that is held collectively and not se

Example Catchment Wisdom statements

1) Embed the Reserve explicitly in operating rules and targets.

- Because: Because SRI evidence shows that ‘rules-in-use’ (e.g., a fixed operational minimum flow) can diverge from the Reserve requirement, leading to recurring non-compliance even when infrastructure is present.

- For whom: Primary adopters: Inkomati CMA operations teams, dam operators, transboundary operating bodies.

- Conditions for success: Conditions: a jointly agreed translation of Reserve requirements into operational targets; routine publication of a monthly compliance brief; annual review based on observed outcomes.

2) Standardise a low-burden transboundary compliance product.

- Because: Because routine, trusted visibility reduces disputes and allows early corrective action before deficits compound.

- For whom: Primary adopters: Inkomati CMA, counterpart agencies in Eswatini and Mozambique; basin forum secretariat.

- Conditions for success: Conditions: one-page monthly summary (targets vs observed; key drivers; actions); shared data definitions and metadata.

3) Manage variability-inducing incentives (e.g., pumping schedules) as a governance issue, not only a technical one.

- Because: Because intra-day and intra-week flow swings can cause ecological stress and reduce the credibility of compliance claims.

- For whom: Primary adopters: irrigator associations, energy utilities, CMA operations, municipalities.

- Conditions for success: Conditions: negotiated revisions to pumping agreements; monitoring of variability and near-zero days; escalation triggers when variability thresholds are exceeded.

Catchment Insights & Wisdoms

8. A cross-catchment synthesis

The Shared Rivers Initiative (SRI) evidence, when viewed across the six Lowveld river systems, shows a consistent pattern: the most persistent implementation barriers are not the absence of technical knowledge, but the translation of that knowledge into day- to- day operating rules, integrated reconciliation planning (water resources + water services), and credible compliance regimes. Across rivers, late dry- season conditions (often around September) amplify stress and expose weak coordination, weak monitoring design, and unclear escalation pathways.

What is context-specific

- Transboundary governance requirements (especially Komati and lower Olifants) and the institutional arrangements that mediate cross-border commitments.
- Catchment-specific hydraulic and operational constraints such as lag times between dam release and downstream response (e.g., Letaba).
- Dominant pressure types: water quality as the primary risk amplifier (Olifants) vs. demand/overallocation (Crocodile, Sand) vs. allocation-before-Reserve challenges (Luvuvhu).
- Existing institutional maturity: systems with established CMA capability and forums can implement faster than systems still in institutional transition.

Possible common products for CCC cross-catchment learning

- Monthly compliance brief: targets vs observed at selected EWR sites; key drivers; actions taken; implications for the next month.
- Seasonal (pre-dry-season) support systems and communication: expected stress level; planned releases/restrictions; priority protection and planning for basic human needs and the Reserve; communication plan.
- Cross-catchment DIKIW Learning Record: what data were used, what was learnt, what insight emerged, what guidance is being adopted, and how outcomes will be tracked.
- Annual 'What travels' synthesis: 5–10 transferable lessons, with clear conditions and adopters, updated through reflection and evidence from the year.

Common indicators to enable comparison across the six catchments

- Reserve compliance (% of days/months failing to meet EWR targets) at agreed 'comparable' sites.
- Frequency of zero or near-zero flow days in late dry season.
- Rule adherence: proportion of time operating rules/releases match planned targets (where rules exist).

- Demand pressure signals: abstraction volumes vs allocation; non-revenue water where municipal demand is a driver.
- Quality pressure signals where relevant: key water quality exceedances and wastewater treatment performance metrics.
- Governance signals: forum activity and follow-through (decisions recorded, actions implemented, outcomes reviewed).

9. Insights and Wisdoms

Insights

- **Reserve non-compliance is widespread and worsening.** None of the rivers assessed meet ecological flow requirements; most have deteriorated over the last decade (Sabie is a partial exception).

- **Progress depends on operationalising IWRM, not just policy.** Where improvement is plausible (e.g., Inkomati/Crocodile), it is linked to catchment-based operating systems that integrate the Reserve, international obligations, and sector targets—supported by stronger regulation.

- **Integration failures are the dominant pattern.** Water resources planning is poorly aligned with municipal water services, sector plans, land-use approvals, and mining expansion—often proceeding without reference to water availability, lawfulness, or cumulative impacts.

- **The Reserve is poorly understood and contested.** Many stakeholders (including municipalities and some regulators) lack practical understanding; the Reserve is often framed as a constraint that benefits "others," undermining buy-in and implementation.

- **Leadership and governance are binding constraints.** Catchments without a clearly empowered lead agent and coherent governance arrangements struggle to act, coordinate, and close feedback loops—especially where roles are unclear or CMA functions are not assigned.

- **Platforms exist, but "talk without action" persists.** Forums and MSPs vary in quality, are often under-resourced, and lack decision mandates, leading to weak accountability and limited collective progress.

- **Multi-scale feedbacks and self-organisation determine performance.** Where WUAs/irrigation boards have functioning monitoring, trusted management, and enforcement tools, regulation works locally—but scaling to catchment level remains weak in many systems.

- **Lawful, regulated systems are prerequisites.** Weak monitoring infrastructure, incomplete data systems, unclear institutional roles, limited legal support, slow licensing, and weak incentives for compliance (especially for municipalities and mining) drive unlawful use and erode Reserve protection.
- **Implementation lags must be managed and tested for reasonability.** Lags are expected in transformation, but several catchments show unacceptable delays requiring benchmarks, indicators, and corrective action.

Wisdoms

Cross-catchment wisdoms for adoption

- **Translate the Reserve into operational targets and rules-in-use.** A determined Reserve only becomes protective when embedded in dam release rules, abstraction schedules, restriction protocols, and routinely verified against observed outcomes.
- **Design monitoring to be decision-ready.** Place gauges, set reporting frequency, and build dashboards around EWR sites and key offtake nodes so compliance assessment directly informs operational control.
- **Integrate water resources management with water services planning to prevent "demand creep."** Municipal expansions, infrastructure upgrades, and WCDM gains must be reconciled with the Reserve and basic human needs, or ecological water is steadily re-absorbed by new demand.
- **Make cooperative governance platforms action-oriented learning infrastructure.** Forums work best when resourced, mandated to synthesise evidence into decisions, and run regular learning-to-action loops (not consultation only).
- **Pair cooperation with a credible compliance backstop.** In high-stress systems, voluntary adherence is not enough—metering/verification, legal literacy, and enforceable escalation steps sustain legitimacy and fairness.
- **Manage incentives that create unintended flow variability.** Where tariffs, pumping schedules, or operational habits create harmful intra-day variability, governance agreements and monitoring must complement technical fixes.
- **Treat water quality and quantity as one risk system in pollution hotspots.** Flow management, wastewater performance, and mining pollution controls must be coordinated—especially where downstream ecological and transboundary/diplomatic risks are significant.
- **Use restoration (e.g., alien invasive clearing) as part of transparent water accounting.** Any "new water" created through restoration should be measured and governed so it

first supports the Reserve and basic human needs before being captured by new demand.

System-wide priorities implied by the evidence

- **Build leadership capacity and mandate clarity** (including functional CMAs where needed) so integrated plans can be executed and feedback loops closed.
- **Harmonise planning instruments** (CMS, WSDPs, PGDS, WCDM, mining and land-use approvals) around water availability, lawful use, and cumulative impacts.
- **Strengthen compliance capacity** (CME skills, legal support, monitoring hardware, incentives) while leveraging capable institutions (WUAs/WSIs) through delegation where appropriate.
- **Set and enforce "lag benchmarks"** so Reserve, licensing, monitoring, and enforcement delays are visible, tested for reasonability, and corrected

10. Sustaining Catchment Wisdom and turning it into action

This report has set out Catchment Wisdom as the CCC's practical bridge between evidence and decision-making. The core message is straightforward: **data and reporting do not automatically produce better outcomes**. What makes the difference is whether partners can move through the full **DIKIW journey**—from data to information, to shared knowledge, to insight, and finally to **collectively agreed guidance** that changes what people do, how institutions coordinate, and what gets prioritised and funded.

The Lowveld case example demonstrates why this matters. Across the six river systems, similar "technical" issues recur—flow stress, water quality hotspots, infrastructure failures, land-use impacts, ecosystem pressures, and competing demands. But the most persistent constraints are often **institutional and governance-related**: how stakeholders interpret evidence, how responsibilities are shared, whether forums are functional, whether rules are implementable, and whether equity is addressed when trade-offs become real. This is why Catchment Wisdom must focus as much on *practice and feasibility* as on measurements. It captures what standard MERL systems often miss: the operational and institutional lessons that determine whether tools like the Reserve, compliance systems, and integrated planning actually work in real conditions.

Catchment Wisdom is also a discipline against "navel-gazing". A wisdom statement is only credible when it is linked to a clear adoption pathway: who must act, what changes in practice, what enabling conditions are required, and when the guidance will be reviewed. This is the difference between a lesson learned and a governance-ready outcome. The Wisdom process therefore needs to be treated as part of the CCC's operating system—built into learning loops, forums, planning cycles, and synthesis moments—rather than as an occasional reflection exercise.

Looking ahead, three priorities stand out for sustaining Wisdom adoption:

1. **Agree the minimum common data points and learning records**
   If each catchment uses a shared template and core indicators, CCC synthesis becomes faster, comparisons become credible, and patterns that are invisible in isolated projects become visible across the collective.

2. **Resource the translation function**
   Wisdom does not "appear" because data exists. It requires convening, facilitation, synthesis, and clear products. Investing in learning labs, cross-catchment exchanges, and short decision-ready briefs is not optional—it is the mechanism that turns evidence into collective guidance.

3. **Measure the wisdom effect, not just the activity**
   The most important success signal is not the number of workshops held or reports produced. It is whether wisdom is adopted into plans, protocols, agreements, budgets, and institutional routines—and whether practice shifts over time. Tracking uptake and change is what makes Catchment Wisdom accountable.

Finally, the purpose of Catchment Wisdom is not only to improve local decisions; it is to strengthen the CCC's ability to influence the wider system. When lessons are synthesised and expressed as clear, evidence-backed guidance, they become actionable for partners such as municipalities and CMAs, and they become credible inputs to research and policy processes (including what is taken forward to the Water Research Commission and other national platforms). In this way, Catchment Wisdom becomes a shared asset: grounded in specific catchments, but powerful precisely because it is **collective, comparable, and transferable**.

The next step is to move from framework to routine practice: establish the annual learning agenda, run Wisdom Labs against real decision windows, produce a small number of high-quality synthesis briefs, and track which wisdom statements are being adopted. If CCC does this consistently, Catchment Wisdom will not be a chapter in a report—it will become a practical capability for adaptive governance, equity-centred decision-making, and resilient catchment management.

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